

Literature Comparison: Plasma Lenses, Space-Charge Compensation, and Proposed Cyclotron Plasma Column

1. Mechanism

- **Plasma/Gabor lenses:** Create an electron cloud by external discharge/ECR source; electron plasma confined by solenoid + electrodes; used as active focusing elements.
- **LEBT SCC (space-charge compensation):** Beam-ionized residual gas generates plasma; electrons partially cancel proton/ H^+ beam space charge; solenoid guides electrons, end-caps set losses.
- **Neutralized drift lines (NDCX, etc.):** Beam propagates in pre-formed plasma at higher pressures (10^{-4} – 10^{-3} torr); nearly full neutralization allows bunch compression.
- **Proposed concept:** Self-ionized plasma column inside axial-injection solenoid, operated at $\sim 10^{-5}$ torr H_2 (with Kr seeding option). Electrode biases (~ 10 – 60 V) confine electrons, vent ions; goal is steady neutralization, not focusing.

2. Gas and Pressure

- **LEBT SCC:** H_2 or N_2 , typical pressures 10^{-6} – 10^{-5} torr; steady SCC fractions ~ 70 – 95% .
- **Plasma/Gabor lenses:** operate with external plasma injection; not UHV-limited; often higher densities.
- **Neutralized drifts:** 10^{-4} – 10^{-3} torr; not compatible with downstream RF cavities.
- **Proposed:** Residual H_2 at 10^{-5} torr baseline; Kr seeding ($\sim \text{few} \times 10^{-6}$ torr partial) to accelerate neutralization. Differential pumping isolates cyclotron tank ($\leq 10^{-7}$ torr).

3. Magnetic Field and Geometry

- **LEBT SCC:** Solenoids 0.05–0.3 T over 0.2–0.5 m typical.
- **Gabor lenses:** Solenoids 0.1–0.4 T; strong electron confinement.
- **Proposed:** 0.1–0.2 T, length 0.1–0.3 m, aperture ~ 30 mm; matched to existing cyclotron axial optics.

4. Neutralization Time Constants (τ)

- **LEBT SCC:** $\tau(H_2, 20\text{--}50 \text{ keV}, 10^{-5} \text{ torr}) \sim 0.05\text{--}0.2 \text{ ms}$.
- **With heavier gases (Ar/Kr):** τ shorter by $\times 3\text{--}5$ due to higher ionization cross sections.
- **Proposed:** For 30 keV, 10 mA p^+ : $\tau \approx 0.064 \text{ ms}$ (H_2), $\tau \approx 0.013 \text{ ms}$ (Kr). Kr improves turn-on for pulsed beams.

5. Scattering/Emittance Growth

- **H_2 :** Minimal scattering at 10^{-5} torr over 0.2 m.
- **Kr:** Larger multiple scattering; should be minimized to transient/pulsed operation.
- **Proposed:** Use Kr only during startup/tuning; H_2 steady operation.

6. Location in Beamline

- **LEBT SCC:** Typically entire LEBT between source and RFQ.
- **Plasma/Gabor lenses:** Used as discrete focusing elements.
- **Proposed:** Drop-in plasma column immediately upstream of cyclotron spiral inflector; compact module with local pumping.

7. Novelty / Distinction

- First integration of SCC cell tailored for **cyclotron axial injection**.
- Explicit **Kr seeding strategy** for fast neutralization without high base pressure.
- **Compact, low-voltage electrode confinement** rather than strong electrostatic lensing.

8. Diagnostics and Validation (from literature → proposal)

- **Emittance scans** before/after plasma cell.
- **Time-resolved current profiles** to measure neutralization build-up.
- **Pressure/Bias/B-field scans** to optimize.

Summary: Prior work demonstrates plasma lenses and SCC in LEBTs and neutralized drift lines. The proposed plasma column for cyclotron injection uniquely combines residual-gas SCC physics with Kr-assisted fast start-up and compact integration ahead of a spiral inflector, under stricter vacuum constraints.